

Day 6 → 06/11/25

Loops → They are used to execute a block of code repeatedly until certain condition is met.
Python mainly have two

Python mainly have two types of loops.

- 1) for loop
- 2) while loop
- 3) Nested loop

1) A for loop is used when you want to repeat a block of code a specific number of times or iterate through items in a sequence.
[like a list, string, tuple, range, etc.]

Example :-

1) Loop through a range of numbers. @.

for loop with range.

```
for i in range(5):  
    print(i)
```

Out put :-

0
1
2
3
4

2) Loop through a sequence @

for loop with sequence

```
fruits = ["apple", "banana", "cherry"]
```

```
for item in fruits:  
    print (item)
```

2) While loop

It is used when you don't know the exact no. of repetitions and continues until a condition becomes false.

```
Ex: count = 1
```

```
while count <= 5:
```

```
    print(count)
```

```
    count += 1
```

3) Nested loop

A nested loop means placing one loop within another loop.

* The inner loop runs completely every time & outer loop runs once.

```
Ex: for i in range(3):
```

```
    for j in range(2):
```

```
        print(i, j)
```

Example problems

```

ol for i in range(1, 11, 1): # i = 1, 2, 3, ... 10
    print(i, " = python")

```

Output 1 = python

1

1

1

10 = python.

```
02 for i in range(1, 11, 1):
    print(i)
```

Output	1
	2
	3
	1
	1
	1
	10

03 print table of given num from n# 1 to n# 2 = 10

$$n^2 = 2 = 2 \times 1$$
$$n \# 10 = a_n h$$

```
n = int(input("enter a number"))
for i in range(1, 11, 1):
    print(f"{n} * {i} = {n*i}")
```